ARJUNA JEE 2027

Lecture -08 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A **Formula's for Multiple & Submultiple Angles**

B **Question Practice**

C

D





LAST CLASS REVISION

Sum or Difference into Multiplication:

$$\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$\sin C - \sin D = 2\sin\left(\frac{C-D}{2}\right)\cos\left(\frac{C+D}{2}\right)$$

$$\cos C + \cos D = 2\cos\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

$$\cos C - \cos D = -2\sin\left(\frac{C+D}{2}\right)\sin\left(\frac{C-D}{2}\right)$$

Multiplication into Sum or Difference :

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

$$2\cos A\sin B = \sin(A+B) - \sin(A-B)$$

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$



LAST CLASS REVISION

$$\tan(A+B+C)=\frac{\tan A+\tan B+\tan C-\tan A\tan B\tan C}{1-(\tan A\tan B+\tan A\tan C+\tan B\tan C)}$$

$$=\frac{S_1-S_3}{1-S_2}$$

$B \to (-B)$

$$\tan(A-B+C)=\frac{\tan A-\tan B+\tan C+\tan A\tan B\tan C}{1-(-\tan A\tan B+\tan A\tan C-\tan B\tan C)}$$

Question Practice

Question

[Ans. : C, D]

If $\prod_{k=1}^{45}(1+\tan k^\circ) = 2^t$, then '$t$' is/are -

A Composite number

B Even number

C Prime number

D Odd number

$$(1+\tan 1^\circ)\times(1+\tan 2^\circ)\times \ldots\ldots\ldots (1+\tan 44^\circ)\times(1+\underbrace{\tan 45^\circ}_{1})$$

22 Times

$$\boxed{2\times 2\times \ldots\ldots 2}\times 2 = 2^{23}$$

Question

If α in first quadrant = $\frac{\pi}{19}$. Find value of $\frac{\sin 23\alpha - \sin 3\alpha}{\sin 16\alpha + \sin 4\alpha}$.

$$\frac{2\sin 10\alpha \cdot \cos 13\alpha}{2\sin 10\alpha \cdot \cos 6\alpha} = \frac{\cos\left(\frac{13\pi}{19}\right)}{\cos\left(\frac{6\pi}{19}\right)} = -1$$

$$\cos\frac{6\pi}{19} = \cos\left(\pi - \frac{13\pi}{19}\right)$$

$$= -\cos\frac{13\pi}{19}$$

$$\frac{13\pi}{19} + \frac{6\pi}{19} = \pi$$

if $A + B = \pi$

$\{\cos A = -\cos B$, $\sin A = \sin B\}$

Question

Prove that $\frac{\sin A + 2\sin 3A + \sin 5A}{\sin 3A + 2\sin 5A + \sin 7A} = \frac{\sin 3A}{\sin 5A}$.

Done !!

Question [JEE-1984]

$$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$$

A $\frac{1}{2}$

B $\cos\frac{\pi}{8}$

C $\frac{1}{8}$

D $\frac{1+\sqrt{2}}{2\sqrt{2}}$

$\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right) \times \left(1-\cos\frac{3\pi}{8}\right)\left(1-\cos\frac{\pi}{8}\right)$

$\left(1-\cos^2\frac{\pi}{8}\right) \times \left(1-\cos^2\frac{3\pi}{8}\right)$

$\sin^2\frac{\pi}{8} \times \sin^2\frac{3\pi}{8}$

$\# \cos\frac{7\pi}{8} = \cos\left(\pi-\frac{\pi}{8}\right)$

$= -\cos\frac{\pi}{8}$

$\left(\frac{1}{2}\cdot 2\sin\frac{\pi}{8}\cdot\sin\frac{3\pi}{8}\right)^2$

$\left(\frac{1}{2}\left(\cos\left(\frac{\pi}{8}-\frac{3\pi}{8}\right)-\cos\left(\frac{\pi}{8}+\frac{3\pi}{8}\right)\right)\right)^2$

$\left(\frac{1}{2}\left(\left(\cos\frac{\pi}{4}\right)-0\right)\right)^2 = \left(\frac{1}{2}\times\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{8}$

Question [JEE-1984]

If $\cos\alpha + \cos\beta = \frac{1}{2}$ & $\sin\alpha + \sin\beta = \frac{1}{3}$, then find $\cos\left(\frac{\alpha+\beta}{2}\right)$, $\cos\left(\frac{\alpha-\beta}{2}\right)$, $\tan\left(\frac{\alpha+\beta}{2}\right)$ & $\tan\left(\frac{\alpha-\beta}{2}\right)$.

θ

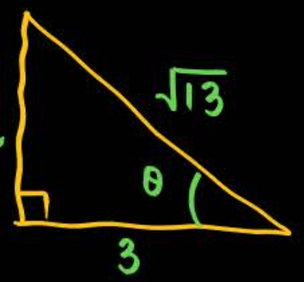
$\frac{2}{3}$

$$\frac{2\cos\left(\frac{\alpha+\beta}{2}\right)\cdot\cos\left(\frac{\alpha-\beta}{2}\right)}{2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right)} = \frac{\frac{1}{2}}{\frac{1}{3}}$$

$$\cot\left(\frac{\alpha+\beta}{2}\right) = \frac{3}{2}$$

I OR III

$\tan\theta = \frac{2}{3}$



$\cos\theta = \pm\frac{3}{\sqrt{13}}$

$\cos\left(\frac{\alpha+\beta}{2}\right) = \pm\frac{3}{\sqrt{13}}$

$\cos(\alpha+\beta) = \pm\frac{6}{\sqrt{13}}$ ✗

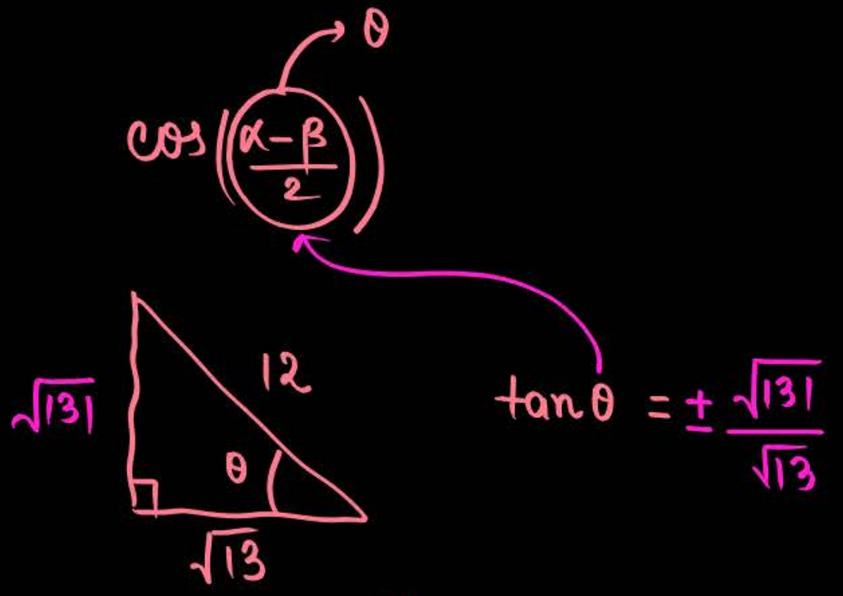
Question [JEE-1984]

If $\cos\alpha + \cos\beta = \frac{1}{2}$ & $\sin\alpha + \sin\beta = \frac{1}{3}$, then find $\cos\left(\frac{\alpha+\beta}{2}\right)$, $\cos\left(\frac{\alpha-\beta}{2}\right)$, $\tan\left(\frac{\alpha+\beta}{2}\right)$ & $\tan\left(\frac{\alpha-\beta}{2}\right)$.

$$2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) = \frac{1}{2}$$

$$2\left(\frac{\pm 3}{\sqrt{13}}\right) \times \cos\left(\frac{\alpha-\beta}{2}\right) = \frac{1}{2}$$

$$\cos\left(\frac{\alpha-\beta}{2}\right) = \pm\frac{\sqrt{13}}{12}$$



$$\cos(\alpha-\beta) = \cos\left(2\left(\frac{\alpha-\beta}{2}\right)\right) = 2\cos^2\theta - 1$$

$$= 2\left(\frac{13}{144}\right) - 1 = -\frac{59}{72}$$

Question [JEE-1984]

If $\cos\alpha + \cos\beta = \frac{1}{2}$ & $\sin\alpha + \sin\beta = \frac{1}{3}$, then find $\cos\left(\frac{\alpha+\beta}{2}\right)$, $\cos\left(\frac{\alpha-\beta}{2}\right)$, $\tan\left(\frac{\alpha+\beta}{2}\right)$ & $\tan\left(\frac{\alpha-\beta}{2}\right)$.

SBS.

$\cos^2\alpha + \cos^2\beta + 2\cos\alpha\cos\beta = \frac{1}{4}$

SBS

$\sin^2\alpha + \sin^2\beta + 2\sin\alpha\sin\beta = \frac{1}{9}$

add

$1 + 1 + 2(\cos\alpha\cos\beta + \sin\alpha\sin\beta) = \frac{1}{4} + \frac{1}{9} = \frac{13}{36}$.

$2\cos(\alpha-\beta) = \frac{13}{36} - 2 = \frac{-59}{36} \Rightarrow \boxed{\cos(\alpha-\beta) = \frac{-59}{72}}$

\# $\cos(\alpha+\beta) = 5/13$

\# $\cos(\alpha-\beta) = \frac{-59}{72}$

$\cos\left(2\left(\frac{\alpha+\beta}{2}\right)\right) = \cos(2\theta)$

$= 2\cos^2\theta - 1 = 2\left(\frac{\pm 3}{\sqrt{13}}\right)^2 - 1 = 2\times\frac{9}{13} - 1$

$= 5/13$.

FORMULAS FOR MULTIPLE & SUBMULTIPLE ANGLES

$A \xrightarrow{\text{multiples}} 2A, 3A, 4A \ldots\ldots$

$\text{sub-multiples} \longrightarrow \frac{A}{2}, \frac{A}{3}, \frac{A}{4} \ldots\ldots$

(1) $\sin(2\theta) = \sin(\theta+\theta)$

$= \sin\theta\cos\theta + \cos\theta\sin\theta$

$= \boxed{2\sin\theta\cos\theta}$

(2) $\cos(2\theta) = \cos(\theta+\theta)$

$= \cos\theta\cos\theta - \sin\theta\sin\theta$

$= \boxed{\cos^2\theta - \sin^2\theta}$

$= \cos^2\theta - (1-\cos^2\theta) = \boxed{2\cos^2\theta - 1}$

$= (1-\sin^2\theta) - \sin^2\theta = \boxed{1-2\sin^2\theta}$

(3) $\tan(2\theta) = \tan(\theta+\theta)$

$= \frac{\tan\theta + \tan\theta}{1 - \tan\theta\tan\theta}$

$= \frac{2\tan\theta}{1-\tan^2\theta}$

FORMULAS FOR MULTIPLE & SUBMULTIPLE ANGLES

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1. $\sin 2\theta = 2\sin\theta\cos\theta$

2. $\cos 2\theta = \cos^2\theta - \sin^2\theta$
$= 2\cos^2\theta - 1$
$= 1 - 2\sin^2\theta$

3. $\left(\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}\right)$

$$\Rightarrow \sin 2\theta = \left(\frac{2\tan\theta}{1+\tan^2\theta}\right)$$

$$\Rightarrow \cos 2\theta = \left(\frac{1-\tan^2\theta}{1+\tan^2\theta}\right)$$

$$\Rightarrow \tan^2\theta = \left(\frac{1-\cos 2\theta}{1+\cos 2\theta}\right)$$

4. $(\sin\theta + \cos\theta)^2 - 1 = \sin 2\theta = 1 - (\sin\theta - \cos\theta)^2$

Also Note

1. $\frac{1+\cos 2\theta}{\sin 2\theta} = (\cot\theta)$

$\frac{1+2\cos^2\theta-1}{2\sin\theta\cdot\cos\theta} = \cot\theta$.

2. $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$

$\frac{1-(1-2\sin^2\theta)}{2\sin\theta\cos\theta} = \frac{2\sin^2\theta}{2\sin\theta\cos\theta} = \tan\theta$.

3. $\frac{1+\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}+\theta\right)$

4. $\frac{1-\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}-\theta\right)$

$\theta \to -\theta$

$\frac{1+2\sin\theta\cos\theta}{\cos^2\theta-\sin^2\theta} = \frac{(\sin\theta+\cos\theta)^2}{(\cos\theta-\sin\theta)(\cos\theta+\sin\theta)} = \frac{\cos\theta\left(\frac{\sin\theta}{\cos\theta}+1\right)}{\cos\theta\left(1-\frac{\sin\theta}{\cos\theta}\right)}$

$= \frac{\tan\theta+1}{1-\tan\theta}$

$= \tan\left(\frac{\pi}{4}+\theta\right)$

Also Note

1. $\frac{1+\cos 2\theta}{\sin 2\theta} = (\cot\theta)$

2. $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$

3. $\frac{1+\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}+\theta\right)$

4. $\frac{1-\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}-\theta\right)$

$\theta \to \frac{\pi}{4}+\theta$

$$\text{LHS} = \frac{1-\cos 2\left(\frac{\pi}{4}+\theta\right)}{\sin 2\left(\frac{\pi}{4}+\theta\right)}$$

$$= \frac{1-\cos\left(\frac{\pi}{2}+2\theta\right)}{\sin\left(\pi/2+2\theta\right)} = \frac{1-(-\sin 2\theta)}{\cos 2\theta}$$

Prove that:

(1) $\sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$

$$RHS = \frac{\left(\frac{2\sin\theta}{\cos\theta}\right)}{1+\frac{\sin^2\theta}{\cos^2\theta}} = \frac{2\sin\theta}{\cos\theta\left(\frac{\cos^2\theta+\sin^2\theta}{\cos^2\theta}\right)} \quad (\cos^2\theta+\sin^2\theta = 1)$$

$$= 2\sin\theta\cos\theta$$

$$= \sin 2\theta .$$

(2) $$\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{1-\frac{\sin^2\theta}{\cos^2\theta}}{1+\frac{\sin^2\theta}{\cos^2\theta}}$$

$$= \frac{\cos^2\theta-\sin^2\theta}{\cos^2\theta+\sin^2\theta} = \cos 2\theta .$$

$(\cos^2\theta+\sin^2\theta = 1)$

$$\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$$

$$(\sin\theta + \cos\theta)^2 = \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta$$

$$= 1 + \sin 2\theta$$

$$(\sin\theta - \cos\theta)^2 = \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$$

$$= 1 - \sin 2\theta$$

$$1 - (\sin\theta - \cos\theta)^2 = \sin 2\theta = (\sin\theta + \cos\theta)^2 - 1$$

$(1 - 2\sin^2\theta)$

$1 - \cos 2\theta = 2\sin^2\theta$

$1 + \cos 2\theta = 2\cos^2\theta$

$2\cos^2\theta - 1$

• S^4

$\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta$.

$\sin 2\theta = 2\sin\theta\cos\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$ / $2\cos^2\theta - 1$ / $1 - 2\sin^2\theta$

$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.

$\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta}$

$\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta}$

OR

$\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$.

$\frac{1+\cos 2\theta}{\sin 2\theta} = \cot\theta$.

$\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$.

** $\frac{1+\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}+\theta\right)$

$\frac{1-\sin 2\theta}{\cos 2\theta} = \tan\left(\frac{\pi}{4}-\theta\right)$

RAPID FIRE

(1) $\sin\frac{\pi}{8}\cdot\cos\frac{\pi}{8} = \frac{1}{2}\sin\left(2\times\frac{\pi}{8}\right) = \frac{1}{2\sqrt{2}}$ ✓

(2) $\left(\cos 120^\circ\right) + 1 = \cos(2\times 60^\circ) + 1$

$= 2\cos^2 60^\circ - 1 + 1 = 2\times\left(\frac{1}{2}\right)^2$

$= \frac{1}{2}$.

(3) $\cos^2\left(\frac{\pi}{12}\right) - \sin^2\left(\frac{\pi}{12}\right) = \cos\frac{2\pi}{12}$

$= \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$.

(4) $\sin^2\frac{\pi}{8} = \frac{1-\cos 2\left(\frac{\pi}{8}\right)}{2} = \frac{1-\frac{1}{\sqrt{2}}}{2} = \frac{\sqrt{2}-1}{2\sqrt{2}}$

$\cos 2\theta = 1 - 2\sin^2\theta$

$\sin^2\theta = \frac{1-\cos 2\theta}{2}$

(5) if $\tan\theta = 2$ then $\tan 2\theta = ?$

$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{2\times 2}{1-4} = \frac{-4}{3}$

(6) if $\tan 2\theta = 2$ then $\tan\theta = ?$

(7) if $\tan\theta = 2$ then $\sin 2\theta = ?$

$\cos 2\theta = ?$

RAPID FIRE

⑥ if $\tan 2\theta = 2$ then $\tan\theta = ?$

$$\frac{2\tan\theta}{1-\tan^2\theta} = 2.$$

$$\tan\theta = t.$$

$$\frac{2t}{1-t^2} = 2^1$$

$$t = 1-t^2$$

$$t^2+t-1 = 0$$

$$\hookrightarrow t = \tan\theta = \frac{-1\pm\sqrt{1-4(-1)}}{2} = \frac{-1\pm\sqrt{5}}{2}$$

⑦ if $\tan\theta = 2$ then $\sin 2\theta = ?$, $\cos 2\theta = ?$

$$\sin 2\theta = \frac{2\tan\theta}{1+\tan^2\theta} = \frac{2\times 2}{1+4} = \frac{4}{5}$$

$$\cos 2\theta = \frac{1-\tan^2\theta}{1+\tan^2\theta} = \frac{1-4}{1+4} = \frac{-3}{5}$$

• S⁴

$\sin 2\theta = 2 \sin\theta \cos\theta$

$\cos 2\theta = \cos^2\theta - \sin^2\theta$ / $2\cos^2\theta - 1$ / $1 - 2\sin^2\theta$

$\tan 2\theta = \dfrac{2\tan\theta}{1 - \tan^2\theta}$.

$\sin\theta = \sin\left(2\dfrac{\theta}{2}\right) = 2\sin\dfrac{\theta}{2}\cos\dfrac{\theta}{2}$

$\sin\dfrac{\theta}{2} = 2\sin\dfrac{\theta}{4}\cos\dfrac{\theta}{4}$.

$\cos 3\theta = \cos 2\left(\dfrac{3\theta}{2}\right) = 2\cos^2\dfrac{3\theta}{2} - 1$.

$\tan\theta = \tan\left(2\dfrac{\theta}{2}\right) = \dfrac{2\tan\theta/2}{1 - \tan^2\theta/2}$

$$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$$

$\theta = \frac{\pi}{8}$

$$\tan\frac{\pi}{4} = \frac{2\tan\frac{\pi}{8}}{1-\tan^2\frac{\pi}{8}}$$

$$1 = \frac{2t}{1-t^2} \Rightarrow 1-t^2 = 2t$$

$$t^2+2t-1=0$$

$$t = \frac{-2 \pm \sqrt{4-4(-1)}}{2}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2}$$

$-1+\sqrt{2} = \tan\frac{\pi}{8}$ ✓

$-1-\sqrt{2} = \tan\frac{\pi}{8}$ ✗

$\tan\frac{\pi}{8} = \sqrt{2}-1$

$\frac{\pi}{4} = 45^\circ$

$\frac{\pi}{8} = (22.5)^\circ$ ← (I)

Question

If $\tan\theta = \frac{3}{4}, \pi < \theta < \frac{3\pi}{2}$, find value of $\sin\frac{\theta}{2}, \cos\frac{\theta}{2}, \cos 2\theta, \tan 2\theta$

III quad.



Acidic Question [JEE Mains]

If $\cot\left(\frac{\pi}{24}\right) = \sqrt{a} + \sqrt{b} + \sqrt{c} + \sqrt{d}$, where $(a, b, c, d \in N)$, then $a + b + c + d =$

$$\tan 2\theta = \frac{2\tan\theta}{1 - \tan^2\theta}$$

$\theta = \pi/24$ or $7.5°$

$$\# \quad 2 - \sqrt{3} = \tan\frac{\pi}{12} = \frac{2\tan\left(\frac{\pi}{24}\right)}{1 - \tan^2\left(\frac{\pi}{24}\right)}$$

$\frac{\pi}{12}$ → $15°$

$$\Rightarrow \quad 2 - \sqrt{3} = \frac{2 \times \frac{1}{\cot\left(\frac{\pi}{24}\right)}}{1 - \frac{1}{\cot^2\frac{\pi}{24}}}$$

$$\# \quad \frac{\pi}{12} = 15° \quad (45° - 30°)$$

half

$$\# \quad \frac{\pi}{24} = (7.5)°$$

$$2-\sqrt{3} = \frac{2 \times \frac{1}{\cot\left(\frac{\pi}{24}\right)}}{1-\frac{1}{\cot^2\frac{\pi}{24}}}$$

$\cot\frac{\pi}{24} = t$.

$$2-\sqrt{3} = \frac{\left(\frac{2}{t}\right)}{1-\left(\frac{1}{t^2}\right)} = \frac{2}{t\left(\frac{t^2-1}{t^2}\right)} = \frac{2t}{t^2-1} = 2-\sqrt{3}$$

$$\frac{2t}{2-\sqrt{3}} = t^2-1$$

$$\frac{1}{2-\sqrt{3}} \times \frac{2+\sqrt{3}}{2+\sqrt{3}} = 2+\sqrt{3}.$$

$$\frac{2t}{2-\sqrt{3}} = t^2 - 1$$

$$2t\left(2+\sqrt{3}\right) = t^2 - 1$$

$$\boxed{t^2 - 2\left(2+\sqrt{3}\right)t - 1 = 0}$$

$$\sqrt{8+4\sqrt{3}} = \left(8+4\sqrt{3}\right)^{\frac{1}{2}} = \left(2\left(4+2\sqrt{3}\right)\right)^{\frac{1}{2}} = \left(2\left(\sqrt{3}+1\right)^2\right)^{\frac{1}{2}} = \sqrt{2}\left(\sqrt{3}+1\right)$$

$$\# \cot\frac{\pi}{24} = t = \left(2+\sqrt{3}\right) \pm \sqrt{\left(2+\sqrt{3}\right)^2 - (-1)}$$

$$= 2+\sqrt{3} \pm \sqrt{4+3+4\sqrt{3}+1}$$

$$= 2+\sqrt{3} \pm \sqrt{8+4\sqrt{3}}$$

$$= 2+\sqrt{3} \pm \sqrt{2}\left(\sqrt{3}+1\right) = 2+\sqrt{3}+\sqrt{6}+\sqrt{2}$$

$$= \sqrt{4}+\sqrt{3}+\sqrt{6}+\sqrt{2} \checkmark$$

If $\cot\left(\frac{\pi}{24}\right) = \sqrt{a} + \sqrt{b} + \sqrt{c} + \sqrt{d}$, where $(a, b, c, d \in N)$, then $a + b + c + d =$

$$\cot\theta = \frac{1+\cos 2\theta}{\sin 2\theta} \xrightarrow{\theta = \frac{\pi}{24}} \cot\frac{\pi}{24} = \frac{1+\cos\frac{\pi}{12}}{\sin\frac{\pi}{12}} = \frac{1+\cos 15^\circ}{\sin 15^\circ} = \frac{1+\frac{\sqrt{3}+1}{2\sqrt{2}}}{\frac{\sqrt{3}-1}{2\sqrt{2}}}$$

$$= \frac{2\sqrt{2}+\sqrt{3}+1}{\sqrt{3}-1} \times \frac{\sqrt{3}+1}{\sqrt{3}+1}$$

$$\sqrt{6}+\sqrt{4}+\sqrt{3}+\sqrt{2} = \frac{2\sqrt{6}+4+2\sqrt{3}+2\sqrt{2}}{2} = \frac{2\sqrt{6}+3+\sqrt{3}+2\sqrt{2}+\sqrt{3}+1}{2}$$

Acidic Question

Show that $\sqrt{2+\sqrt{2+\sqrt{2+2\cos 8\theta}}} = 2\cos\theta$

$2(1+\cos 8\theta)$

$2(1+2\cos^2 4\theta - 1)$

Question [JEE Mains-2020]

If $\frac{\sqrt{2}\sin\alpha}{\sqrt{1+\cos 2\alpha}} = \frac{1}{7}$ and $\sqrt{\frac{1-\cos 2\beta}{2}} = \frac{1}{\sqrt{10}}$, $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$ then $\tan(\alpha + 2\beta)$ is equal to

Hw.

FORMULAS FOR TRIPLE ANGLE

Formula for (3θ) :

1. $\sin(3\theta) = 3\sin\theta - 4\sin^3\theta$
2. $\cos(3\theta) = 4\cos^3\theta - 3\cos\theta$
3. $\tan(3\theta) = \left(\frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}\right)$



$$\# \quad \sin(3\theta) = \sin(\overset{A}{2\theta} + \overset{B}{\theta})$$

$$= \sin 2\theta \cos\theta + \cos 2\theta \sin\theta$$

$$= 2\sin\theta\cos\theta\cos\theta + (1 - 2\sin^2\theta)\sin\theta$$

$$= 2\sin\theta(1 - \sin^2\theta) + (1 - 2\sin^2\theta)\sin\theta$$

$$= 2\sin\theta - 2\sin^3\theta + \sin\theta - 2\sin^3\theta$$

$$\# \quad \boxed{\sin(3\theta) = 3\sin\theta - 4\sin^3\theta}$$



IMPORTANT VALUES TO MEMORISE

Important Values to Memories :

(1) $\sin 18^\circ = \frac{\sqrt{5}-1}{4} = \cos 72^\circ$

(2) $\sin 54^\circ = \frac{\sqrt{5}+1}{4} = \cos 36^\circ$

(3) $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ$

(4) $\sin 72^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4} = \cos 18^\circ$

(5) $\tan\left(\frac{\pi}{8}\right) = \sqrt{2}-1 = \cot\left(\frac{3\pi}{8}\right)$

(6) $\tan\left(\frac{3\pi}{8}\right) = \sqrt{2}+1 = \cot\left(\frac{\pi}{8}\right)$

$\theta = 18^\circ$

$5\theta = 90^\circ$

$3\theta + 2\theta = 90^\circ$

$3\theta = 90^\circ - 2\theta$

$\cos 3\theta = \cos(90^\circ - 2\theta)$

$4\cos^3\theta - 3\cos\theta = \sin 2\theta$

$4\cos^2\theta - 3\cos\theta = 2\sin\theta\cos\theta$

$4(1-\sin^2\theta) - 3 = 2\sin\theta$

$\sin\theta = t$

$4 - 4t^2 - 3 = 2t$

$4t^2 + 2t - 1 = 0$

$$t = \frac{-2 \pm \sqrt{4 - 4(-1)4}}{4 \times 2}$$

$$= \frac{-2 \pm 2\sqrt{5}}{8} = \frac{-1 \pm \sqrt{5}}{4}$$

$\sin(18^\circ) = t = \frac{\sqrt{5}-1}{4}$ ✓ , ~~$\frac{-\sqrt{5}-1}{4}$~~

DO IT BY YOURSELF (DIBY)

DIBY-23

Prove that

1. $\frac{\sin 7\theta-\sin 5\theta}{\cos 7\theta+\cos 5\theta}=\tan\theta$

2. $\frac{\cos 6\theta-\cos 4\theta}{\sin 6\theta+\sin 4\theta}=-\tan\theta$

3. $\frac{\sin A+\sin 3A}{\cos A+\cos 3A}=\tan 2A$

4. $\frac{\sin 7A-\sin A}{\sin 8A-\sin 2A}=\cos 4A\sec 5A$

5. $\frac{\cos 2B+\cos 2A}{\cos 2B-\cos 2A}=\cot(A+B)\cot(A-B)$

6. $\frac{\sin 2A+\sin 2B}{\sin 2A-\sin 2B}=\frac{\tan(A+B)}{\tan(A-B)}$

7. $\frac{\sin A+\sin 2A}{\cos A-\cos 2A}=\cot\frac{A}{2}$

8. $\frac{\sin 5A-\sin 3A}{\cos 3A+\cos 5A}=\tan A$

9. $\frac{\cos 2B-\cos 2A}{\sin 2B+\sin 2A}=\tan(A-B)$

10. $\cos(A+B)+\sin(A-B)=2\sin(45^\circ+A)\cos(45^\circ+B)$

11. $\frac{\cos 3A-\cos A}{\sin 3A-\sin A}+\frac{\cos 2A-\cos 4A}{\sin 4A-\sin 2A}=\frac{\sin A}{\cos 2A\cos 3A}$

DIBY-24

Find the value of expression $\frac{\sin 8\theta\cos\theta - \sin 6\theta\cos 3\theta}{\cos 2\theta\cos\theta - \sin 3\theta\sin 4\theta}$, given $\theta = 7.5°$.

(A) $2+\sqrt{3}$ (B) $2-\sqrt{3}$ (C) 0 (D) 1

[Ans. : (B)]

DIBY-25

If $\alpha = \frac{\pi}{19}$, find the value of $\frac{\sin 23\alpha - \sin 3\alpha}{\sin 16\alpha + \sin 4\alpha}$.

(A) 1 (B) 0 (C) 2 (D) –1

[Ans. : (D)]

DIBY-26

The expression $\frac{(1+\tan 37°)(1+\tan 8°)(1+\tan 15°)(1+\tan 30°)}{(1+\tan 2°)(1+\tan 43°)}$ reduces to

(A) 2 tan 45° (B) 4 (C) 8 (D) 6 cot 45°

[Ans. : (A)]

DIBY-27

For $\alpha, \beta \in \left(0, \frac{\pi}{2}\right)$, let $3\sin(\alpha+\beta) = 2\sin(\alpha-\beta)$ and a real number k be such that $\tan\alpha = k\tan\beta$. Then the value of k is equal to

(A) 5 (B) −2/3 (C) −5 (D) 2/3

[Ans. : (C)]

DIBY-28

Let $\frac{\tan 50^\circ - \tan 40^\circ}{\tan 10^\circ} = k$, find $k \in N$.

[Ans. : 2]



Homework

Re-attempt all Class Examples

DPP & DIBY

Sticky - formula Notes.

It is never about end result

It's always about Journey

Thank You

#futureIITians